



Conference Proceedings

2 Days Conference on Advanced Computational Intelligences Methods and Their Applications

December 27-28, 2023

Kohat, Pakistan.

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Parental Quality Time Investment on Students' Academic Achievement at Secondary School Level

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Abstract: This study investigates the intricate relationship between parental involvement and the academic performance of students, specifically concentrating on the concept of "quality time investment." Our research objectives encompass various dimensions of this relationship. Initially, we seek to elucidate participants' perceptions of "quality time investment" through qualitative methodologies such as interviews, observations, and focused group discussions. These qualitative approaches aim to garner insights into how parents conceptualize and prioritize this notion. Subsequently, employing rigorous statistical analyses, we aim to explore the potential effects of parental quality time investment on students' academic achievements, striving to establish correlations and causal connections. Furthermore, our research objectives encompass determining the optimal level of parental attention required to enhance academic performance, assessing the impact of regularity and consistency in parental practices, and investigating the consequences of avoiding multitasking during educational interactions. Our findings emphasize the critical significance of cultivating a strong parent-child bond to positively influence academic success and overall well-being. Many parents express concerns about their children's academic progress but may face challenges in providing effective support, resulting in stressful interactions. This research carries significant implications for educators, policymakers, and parents. It highlights the crucial necessity for collaborative efforts aimed at promoting effective parental involvement in education. We recommend programs emphasizing unconditional affection, parental patience, and child-centered academic support. Prioritizing these strategies holds the potential to collectively enhance children's academic success and overall well-being.

Perturbative Analysis Of Kubo Conductivity In Topological Insulator In Proximity To A Helical Spin Density Wave

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Abstract: In this research work, the electrical conductivity of Dirac fermions in topological insulators in proximity to a helical spin density wave is investigated. The Kubo conductivity is determined within the framework of second-order perturbation theory. We find that the conductivity is affected significantly by proximity-induced periodic potential amplitude, in particular, its relative strength of the x-, and z-components. In turn, the effect of temperature and Fermi energy on the Kubo conductivity is studied for various values of these parameters.

Investigating Magnetic Heating Efficiency of Nanoferrites $\text{Ni}_{0.5}\text{Co}_{0.5}\text{Fe}_2\text{-xCu}_x\text{O}_4$ via MATLAB and their use in Cancer Therapy

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Abstract: Magnetic hyperthermia, utilizing heat from magnetic nanoparticles (MNPs) in alternating magnetic fields, is explored in this study. $\text{Ni}_{0.5}\text{Co}_{0.5}\text{Fe}_2\text{-xCu}_x\text{O}_4$ nanoparticles, synthesized via sol-gel auto-combustion and oleic acid-coated through ball milling, are structurally and morphologically analyzed using X-ray diffraction and scanning electron microscopy. Magnetic properties are assessed with a vibrating sample magnetometer, noting reduced saturation magnetization with increased doping. Specific absorption rates (SAR) are calculated from heating curves, generated using a precision induction heater. The maximum SAR for oleic acid-coated nanoparticles is 9, with MATLAB-based comparisons confirming these findings. The study also evaluates nanoparticle toxicity in biological environments through hemolysis percentage assays, indicating lower toxicity and potential application in cancer therapy.

Exploiting Optimal Context Window Size and Classification Models for the Identification of Important Citations in Scientific Literature

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Abstract: In the realm of scientific literature, researchers extensively rely on citation counts for evaluating institutions, peer assessments, and calculating journal impact factors. However, there is a consensus among researchers that not all citations carry the same level of significance and weight. Recognizing the inadequacy of solely relying on citation counts, researchers advocate for a deeper examination of the reasons behind each citation. Various motivations for referencing a particular work have been identified, ranging from presenting background knowledge to building upon the research of other scholars. In response to these considerations, various taxonomies and machine learning models have been developed for citation analysis. Nevertheless, these approaches often fall short in capturing the internal context inherent within these in-text citations. Acknowledging the importance of capturing word semantics, the selection of suitable word embeddings and classification models becomes paramount for achieving superior analytical results. Word embeddings offer n-dimensional distributed representations of text, aiming to capture the nuanced meanings of words. Deep learning-based word embedding techniques have gained prominence in Natural Language Processing (NLP) tasks, including text classification, sentiment analysis, and citation analysis. Existing state-of-the-art techniques frequently operate on small datasets with fixed context window sizes, resulting in the loss of contextual meaning. This study leverages two benchmark datasets with a substantial volume of in-text citations to guide the selection of an optimal content window size and classification approaches. A comparative analysis of various context window sizes for in-text citations is conducted to effectively identify important citations. The study employs Word2Vec embeddings in conjunction with deep learning models (Convolutional Neural Networks, Gated Recurrent Units, Long Short-Term Memory), Support Vector Machine, Decision Tree, and Naive Bayes. Evaluation metrics, including precision, recall, F-score, and accuracy, are employed for each combination of context window sizes. The findings highlight that, particularly for lengthy in-text citations, larger citation windows excel in capturing the semantic essence of citations. In the context of citation analysis tasks, a window size of 10 demonstrates superior accuracy, recall, and precision with both machine and deep learning models.

Enhancing Guava Quality Assessment with Multi-Class Detection using RetinaNet for Ripeness, Bruising, and Disease Identification

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Abstract: The demand for accurate and efficient quality assessment of agricultural produce has surged with the increasing emphasis on food quality and safety. This research paper introduces a novel approach to guava quality assessment through the implementation of the RetinaNet model, a state-of-the-art deep learning framework. The proposed methodology focuses on the simultaneous detection and classification of multiple quality attributes, specifically ripeness, bruising, and disease, contributing to a comprehensive evaluation of guava fruit quality.

Visualizing the Unseen: Supercharged GANs for Video-based Rare Event Detection

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Abstract: Rare event detection in videos poses a substantial challenge, requiring advanced techniques to discern and depict infrequent occurrences. Despite progress in generative modeling, persistent challenges such as mode collapse, training instability, and limited adaptability to diverse real-world scenarios emphasize the need for innovative solutions. ST-WGAN, an evolution of the Spatial-Temporal Wasserstein Generative Adversarial Network (ST-WGAN) framework, introduces a specialized architecture incorporating spatial and temporal generators and discriminators. This integration yields synthesized video frames faithfully replicating real-world scenarios. Tailored loss functions for spatial and temporal discriminators significantly enhance the model's stability and efficacy. Rigorous validation of ST-WGAN on a diverse dataset underscores its superiority in stable training dynamics, diverse sample generation, and high-quality synthetic video content production. Practical efficacy is affirmed through a comprehensive evaluation using metrics such as ROC-AUC and PRC-AUC. The proposed approach's effectiveness for rare event detection is further validated through experiments on the UCSD datasets and UCF Crime dataset, emphasizing its applicability to real-world surveillance scenarios. This research contributes a robust solution to the challenges in rare event detection, enhancing the capabilities of surveillance systems and advancing public safety.

Analysis of Newtonian Heating in Fractional Hybrid nanofluid with the influence of transverse Magnetic Field

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Abstract: In this presentation we will analyzed the flow of flow of Fractional Hybrid nanofluid. Oscillatory wall driven flow is also influenced by Newtonian Heating and a uniform transverse magnetic field. The problem is modelled with generalized Fourier law with the help of Caputo-Fabrizio time Fractional derivative. The system is solved with the of integral transforme namely Laplace transform. The influence of involved Physical parameters is analyzed graphically. It is observed that the role of fractional parameter is more efficient on the flow of hybrid nanofluid.

The Impact of The Changes in Degree Days on Environmental Sustainability in South Asia

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Abstract: Environmental sustainability is needed to limit environmental harm and resource depletion and involves decisions that will affect all living things, natural resources, and the climate in future. Several variables including Cooling Degree Days, Heating Degree Days, GDP, Trade Openness and Urbanization which play crucial positive or negative role in climate change. Environmental sustainability is studied in relation to these variables in selected South Asian countries. The FMOLS model was used to study the relation between environmental sustainability and five selected variables. A negative relation was observed between Cooling degree days and environmental sustainability. Heating Degree days also demonstrated a negative correlation with Environmental sustainability. Our study concluded that serious policies are needed to be materialized so that environmental degradation is avoided especially considering Heating Degree Days, Cooling Degree Days and Urbanization which showed negative relation and will have serious consequences in future.

Honesty Based Watchdog System to Avoid Selfish Behavior of Nodes in Vehicular Delay-Tolerant Networks: An Energy Model

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Abstract: The concept of a Delay Tolerant Network (DTN) emerges as a feasible remedy for addressing challenges encountered by applications dealing with sporadic connectivity, prolonged propagation delays, and elevated error rates. Vehicular Delay-Tolerant Networks (VDTNs) play a pivotal role in enabling reliable and efficient communication among vehicles in challenging and dynamic vehicular environments. However, the presence of selfish behaviour among vehicles can disrupt the seamless operation of VDTNs, leading to inefficient resource utilization and compromised network performance. In this research paper, we introduce an innovative Honest Watchdog System (Watch H) tailored for VDTNs. Social parameters (interaction frequency, credibility and energy) of honesty will be used to select honest watchdog system. Our proposed honest watchdog system Watch_H analyses the energy level of the nodes in the network. The nodes normally show selfishness to save its main energy resources. Unlike many watchdog systems where the selfish nodes are identified and the actions are taken to deal with them. In the proposed model the Watch_H system will use a threshold value of energy to avoid the selfishness before the nodes get selfish, thus improving the overall performance of the network. In this article, an algorithm is designed to avoid the selfish behavior of nodes by evaluating the degree level of energy of nodes.

Modeling Temporal Evolution in Heterogenous Networks using Graph Neural Networks

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Abstract: Graph Neural Networks (GNN) have achieved remarkable attention due to their capability to model complex relationships within graph-structured data. When it comes to evolving temporal relationships in heterogenous networks, the application of GNNs can be expanded to capture temporal dynamics. This study investigates the use of GNNs to model evolving temporal relationships in complex networks within the framework of temporal factors, where relationships change over time. Node classification and edge prediction can be studied for temporal change detection in complex systems represented by heterogenous graph networks. However, the availability of limited labelled data is a significant challenge in studying real world systems for temporal evolution. To this end, we propose using the concept of temporal graphs, representing network structures at distinct time points. Then, GNN is leveraged to integrate temporal aspects into the learning process. The proposed GNN based model is trained using benchmark datasets from real world data. To showcase promising applications in temporal evolution, the efficiency of the proposed GNN based model is assessed in predicting graph evolution using performance evaluation metrics techniques. Furthermore, the challenges with irregular temporal sampling in real world datasets and the applicability of models to large evolving networks are also discussed. We demonstrate the effectiveness of the suggested GNN based model in capturing evolving temporal relationships through experiments and evaluations, providing a potential avenue for understanding spatiotemporal features in heterogenous network structures.

Unveiling Software Flaws And Self-Admitted Technical Debt By Odyssey Of Deep Learning Analytics

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Abstract: Software development teams often prioritize quality over timely and cost-effective completion, leading to the buildup of technical debt (TD) which can be a trade-off between short-term benefits and long-term system health. TD can be categorized into design and code debt, and if not paid off quickly, it can accumulate interest similar to a loan. TD is common and affects software quality, complexity, and maintenance. Research shows that 20-29% of admitted technical debt is in code comments. The proposed system assists software professionals in making informed decisions and taking proactive measures to mitigate SATD and bugs, improving software quality, reducing maintenance costs, and increasing software development efficiency. It aims to address research limitations by integrating a method for identifying Software Acceptable Testing (SATD) and detected bugs, providing a comprehensive classification scheme. It uses deep learning techniques to analyze software code and comments, learning complex patterns, important features (that contribute to SATD) and contextual information.

Numerical Solutions of Global Variable-order Time Fractional Diffusion Type Models Via Non-orthogonal Polynomials

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Abstract: In this work, a hybrid scheme based on Lucas and Fibonacci polynomials is developed for the numerical solutions of one dimensional (1D) linear variable order time fractional partial differential equations (VOTFPDEs). The scheme consisting of a quadrature formula for the fractional part and combined Lucas and Fibonacci polynomials for spatial derivative and solution. Then, with the help of collocation approach the given problem reduces to the system of linear algebraic equations. By solving this system of equations the Lucas series coefficients are obtained. Then, substituting these coefficients into Lucas series expansion approximate solutions can be constructed recursively. Efficiency and performance of the proposed method is examined by calculating L2 and L1 norms. The obtained results demonstrate the usefulness and efficiency of the proposed method.

An Implicit Finite Difference Scheme For The Study Of Local Variable Order Fractional Partial Differential Equations

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Abstract: This thesis presents a numerical scheme based on implicit finite difference scheme for the study of local variable order fractional partial differential equations. The time fractional derivative is considered local variable order sense for both linear and nonlinear problems. First, the temporal derivative will be approximated as product of time dependent function and the usual time derivative. Then, the usual time derivative and space derivatives will be estimated by finite differences. For linear problems the scheme transforms the time fractional models to the system of linear equations. In nonlinear cases Quasilinearization is utilized to tackle the nonlinear parts. The solutions of the system of linear systems gives the solutions of the time fractional models. Next, the stability of the scheme is discussed via Von Neumann method which shows that the scheme is unconditionally stable. Further the performance of the suggested scheme is portrayed with some linear and nonlinear problems which includes Fitzhugh- Nagumo type equations, and coupled of partial differential equations. To confirm the applicability of the method two norms namely, L^∞ and L_2 are calculated. Moreover, the results are addressed in the form of tables and graphs which reveal that the scheme properly works for considered problems.

Automatic Plant Counting And Classification Using Mask R-Cnn And Efficientnet

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Abstract: Plants play a pivotal role in nation's economy across the world and provide basic human needs such as food, clothing, shelter, and health care. The conventional method involves manual counting, typically conducted by botanical experts, which is slow, prone to errors, laborious, and demanding and yields results of lower quality. AI-based methodologies offer resolution by employing advanced machine learning (ML) algorithms and autonomous platforms equipped with sensors to gather data autonomously. ML, a subset of AI, entails training algorithms for specific functionalities using well-annotated datasets, removing the necessity for manual enumeration. However, existing automated plant counting methods also need help achieving high accuracy, particularly when dealing with diverse plant species datasets, variable environmental conditions, and intricate background morphology. Therefore, in this research, image smoothness noise was eliminated by applying color channelization and median filter. Mask R-CNN technique was applied to segment the background and foreground objects and effectively separate occluded objects in UAV images. After segmentation, the plants will be accurately counted and classified with the help of EfficientNet and achieved a higher accuracy than existing state-of-the-art techniques. The final findings of the proposed technique will also be cross-validated by botanical experts.

Novel Parameters Based Trigonometric Probability Distributions with Bounded Support

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Abstract: In this research, new one-parameter unit probability distribution which is defined on trigonometry is proposed. It has been proved that it is more flexible than some existing well-known distributions due to its different shapes of probability density function and hazard rate function. Some of its statistical properties have been studied and explicit form expressions for moments, quantile function, and order statistics are obtained. The method of maximum likelihood is used to estimate the model parameters. The simulation study is conducted for reinforcement of the results. Three real data sets are considered to illustrate the applicability of the proposed model.

Linguistic q-rung orthopair fuzzy Hamacher aggression operator-based MCGDM Analysis

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Abstract: The linguistic q-rung orthopair fuzzy (Lq-ROF) set serves as a useful way for presenting uncertain information by offering more space for decision experts. In the present research, we first link the concepts of Hamacher t-norm and t-conorm with the frame of Lq-ROF numbers to develop and analyze the innovative Lq-ROF Hamacher operations. Then, following the proposed Hamacher's norm operations, a series of aggregation operators including Lq-ROF weighted averaging, Lq-ROF ordered weighted averaging, Lq-ROF hybrid averaging, Lq-ROF weighted geometric, Lq-ROF ordered weighted geometric, Lq-ROF hybrid geometric operators are investigated. Some interesting aspects of these AOs are also proffered. We further develop evaluation based on distance from average solution (EDAS)

approach in light of the newly outlined concepts to cope with Lq-ROF decision-making problems where the weight information of criteria is fully unknown, ultimately, the practicality of the framed approach is demonstrated through an empirical case, and an analysis is carried out to showcase the methodology dominance.

Effects of spacetime perturbation on dark energy, dark matter and dynamics of particle around black hole

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Abstract: The spherical symmetry of heavenly bodies is an important property in gravitational theories. All celestial bodies are approximately spherical shapes due to their gravitational pull in all directions. However, observation shows that the Earth and other planets are not perfect spheres, they are flattened at the poles and bulged at their equators. The black holes are considered to be spherical in shape. Their accelerated motion radiates mass in the form of gravitational waves. It is proved by Hawking that the black holes evaporate in radiation called Hawking radiation. It means that their mass-energy momentum tensor is a decreasing function of time. But all the known black hole solutions are static. To get the time dependent mass-energy momentum tensor we need a time dependent black hole solution of the Einstein field equations. To form the black hole spacetime time dependent, we perturbed the general spherically symmetric spacetime metric by a general time dependent conformal factor. Then use the Noether theorem and obtain a particular time dependent conformal factor without losing the spherical symmetry of the spacetimes produced by heavenly bodies. We define several theorems in this regard, which will provide a way to convert the static form of spherically symmetric static spacetimes (particularly black holes solutions) into non static spacetimes (non static black holes solutions). All spherically symmetric distributions that satisfy certain conditions will admit the time conformal approximation.

Employment Of Computational Intelligence In Sentiment Analysis Of Product Reviews

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Abstract: Sentiment Analysis is the digital orientation of public feelings and opinions shared in textual form. Social media sites are becoming the de-facto skillset for real-world organizations and business companies. Public sentiments and opinions are reshaping the decision-making process in these organizations. Among many social networking websites, public figures as well as common users prefer to use Twitter as their communication platform. Twitter allows only 280 characters to express and post users' sentiments. This limitation brought about inventive challenges to natural language engineers in the form of informal and code-switched text. Online users post feelings, expressions, and sentiments in formal as well as informal and code-switched forms. These sentiments are mined and summarized via supervised and unsupervised machine learning methods, resulting in significant outcomes for resource-rich formal language text. However, contrary to computationally developed languages, resource-poor languages lack real-world analysis due to linguistic complexities and computational resource scarcity. The generic machine translation systems alter the sense of sentiment of scarce resource languages. Likewise, the supervised machine learning systems lack comprehensive coverage of informal terms due to the famine of resources. Deep learning methods and transformers can play a pivotal role in the

semantic classification of resource-poor language. This paper focuses on assessing the ability of computational intelligence to enrich the text resources for computationally less developed languages. To ensure the effectiveness of the proposed study, sentiment analysis of product reviews is performed. Experimental results verified that optimized language processing and BERT considerably improve the overall accuracy.

THE END
